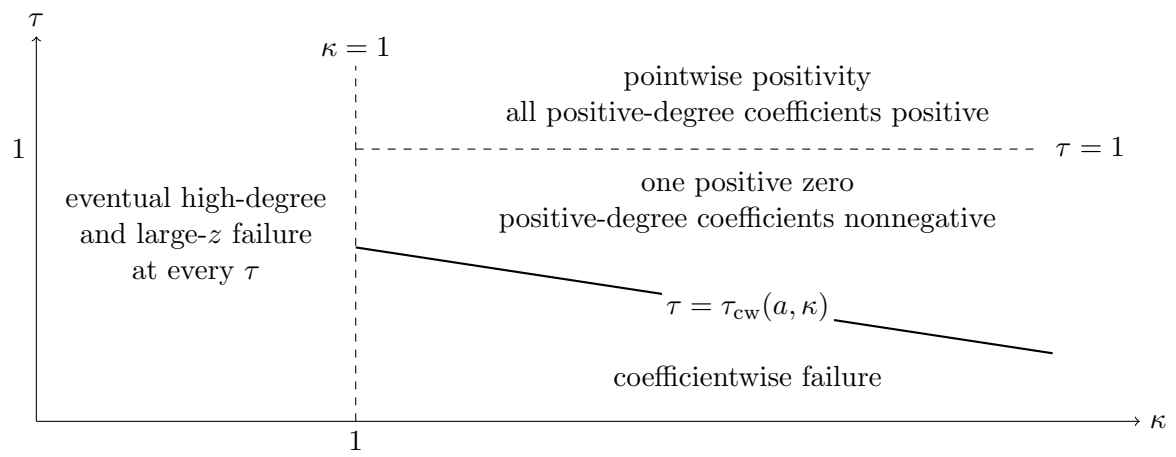




(a)	M_0 $\succeq 0$, rank one	M_1 possibly indefinite	$M_m, m \geq 2$ $\succ 0$
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indefinite for $0 < a < a_*$; $\succeq 0$ of rank one at $a = a_*$;
 $\succ 0$ for $a > a_*$, where $a_* = 0.3690738484\dots$

(b) row Δ_n : the ordered pairs (k, ℓ) with $k + \ell = n$

Δ_1	0, 1	1, 0				
Δ_2	0, 2	1, 1	2, 0			
Δ_3	0, 3	1, 2	2, 1	3, 0		
Δ_4	0, 4	1, 3	2, 2	3, 1	4, 0	
Δ_5	0, 5	1, 4	2, 3	3, 2	4, 1	5, 0
$\vdots$						

- one index = 1: mixed pairing, strictly positive
- the sole self-pair: $\text{MD}(M_1, M_1) = 2 \det M_1$
- neither index 1: both fibers positive semidefinite

